

ORF307: Optimization

Precept: #10

Agenda

- Exercises 3, 6, and 7 from Spring 2023 midterm-2 exam **preparation** (see *AdditionalExercises_02_midterm.pdf* file on Canvas/Modules → "Midterm2 past exams for practice")

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Exercise 3

Let $A = A^T$ be symmetric matrix. Consider the LP

$$\begin{array}{ll} \min_x & c^T x \\ s.t. & Ax \geq c \\ & x \geq 0 \end{array} \quad (1)$$

Show that if x^* satisfies $Ax^* = c, x^* \geq 0$, then x^* is optimal.

Solution:

Convert the original problem to the following equivalent LPs:

$$\begin{array}{ll} \min_x & c^T x \\ s.t. & -Ax \leq -c \\ & x \geq 0 \end{array}$$

$$\begin{array}{ll} \min_x & c^T x \\ s.t. & Ax + s = -c \\ & x, s \geq 0 \end{array}$$

The last LP is in standard form

$$\begin{aligned} \min_x \quad & \tilde{c}^T \tilde{x} \\ \text{s.t.} \quad & \tilde{A} \tilde{x} = \tilde{b} \\ & \tilde{x} \geq 0 \end{aligned} \tag{2}$$

where

$$\tilde{x} = \begin{pmatrix} x \\ s \end{pmatrix}, \quad \tilde{c} = \begin{pmatrix} c \\ 0 \end{pmatrix}, \quad \tilde{b} = -c, \quad \tilde{A} = [-A \ I]$$

Take the dual of (2)

$$\begin{aligned} \max_y \quad & -\tilde{b}^T y \\ \text{s.t.} \quad & \tilde{A}^T y + \tilde{c} \geq 0 \end{aligned}$$

Expand the above problem to get

$$\begin{aligned} \max_y \quad & c^T y \\ \text{s.t.} \quad & -A^T y + c \geq 0 \\ & y \geq 0 \end{aligned}$$

Use the fact that A is symmetric and substitute $A^T = A$

$$\begin{aligned} \max_y \quad & c^T y \\ \text{s.t.} \quad & Ay \leq c \\ & y \geq 0 \end{aligned}$$

But this is the dual of the original problem (1).

NOTE: Recall that a pair (x^*, y^*) is primal/dual optimal iff

- x^* is primal feasible
- y^* is dual feasible
- They make the duality gap zero.

REMARK: The result of this exercise is not only of theoretical interest but it also has practical implications especially in Network Optimization problems. The following example serves this purpose.

Example: *Coordinated Emission Reductions Across U.S. States.*

Suppose all 50 U.S. states form a coalition to combat climate change by reducing greenhouse gas emissions. Their collective goal is to ensure that each state receives a minimum level of environmental benefit, such as cleaner air, improved public health, or enhanced climate resilience, while minimizing the total cost of achieving these benefits.

Each state contributes effort (e.g., through clean energy investments, industrial regulations, or conservation programs) that not only benefits itself but also its direct neighbors due to their shared ecosystems. The **goal** is to determine how much each state should reduce its emissions so that:

- Every state receives at least its required environmental benefit.
- The total cost of effort across all states is minimized.
- The burden of reductions is distributed fairly and efficiently.

To model this as an LP we first define:

- $x \in \mathbb{R}^{50}$ which represents the vector of emission reduction efforts by each state, measured in millions of tons of CO₂ equivalent. Each x_i is a **decision variable** representing how much state i needs to reduce its emissions.
- $c \in \mathbb{R}^{50}$ is the vector of unit costs, where c_i is the cost per million tons of CO₂ reduction for state i . We assume that each state's minimum required environmental benefit is **numerically equal to** its unit cost. Thus, the same vector c appears in both the objective and the right-hand side of the constraint, but with different interpretations.
- $A \in \mathbb{R}^{50 \times 50}$ is the symmetric **influence matrix**, where A_{ij} quantifies how much state j 's reduction benefits state i . We assume $A_{ij} = A_{ji}$ and $A_{ij} = 1$ if states i and j share a border, and $A_{ij} = 0$ otherwise.

The resulting LP can be defined as:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax \geq c \\ & && x \geq 0 \end{aligned}$$

Interpretation of the Constraints: Each row a_i^T of the matrix A represents how all other states' reductions affect state i . The constraint $a_i^T x \geq c_i$ ensures that state i receives at least its required environmental benefit c_i . Since A is symmetric and sparse (nonzero only for neighboring states), the model captures localized mutual influence of neighboring states.

Optimality via Feasibility: Suppose we find a vector $x^* \geq 0$ such that:

$$Ax^* = c$$

Then, by the result of the Exercise 3, discussed earlier, x^* must be optimal. No other feasible allocation of CO₂ reductions can achieve the targets at lower total cost. This is a very useful result where **feasibility with equality and nonnegativity certifies optimality**.

Modeling insights: This example demonstrates how symmetry and alignment between constraints and objectives can arise in models of mutual influence or shared responsibility. By exploiting the structure of the LP allows us to characterize feasible solutions as optimal when the decision variables are non-negative and the constraints are active (i.e., satisfied as equalities).

In addition, this example shows how to model cases where different parties, like the 50 US states, work together to meet common goals. Each state's effort helps others, and the

goal is to find the most cost-effective way to distribute the burden of emission reductions so that everyone benefits.

Of course, this example is a simplification of the very complex environmental problem, and serves just as an example to demonstrate the usefulness of the result we proved earlier and the expressive power of Linear Programming and Mathematical Optimization in general.

Exercise 6

The matrix $P \in \mathbb{R}^{n \times n}$ is a column stochastic if

- $P \geq 0$ (i.e., $p_{ij} \geq 0, \forall i, j$)
- $P^T \mathbf{1} = \mathbf{1}$ (i.e., the column of P sum up to 1, $\sum_{i=1}^n p_{ij} = 1, \forall j$.)

NOTE: Do not confuse the conditions: $P \geq 0$ and $Q \succeq 0$. The former means that all the elements of the matrix P are nonnegative numbers, whereas the later means that the matrix Q is PSD. Can you think of a PSD matrix that contains both negative and nonnegative elements?

In this exercise, there are two tasks for you:

1. Show that if P is column-stochastic then $(P^T x)_i \leq x_{\max}$

Solution:

$$(P^T x)_i = \sum_{j=1}^n P_{ji} x_j \leq \sum_{j=1}^n P_{ji} x_{\max} = x_{\max} \sum_{j=1}^n P_{ji} = x_{\max} \mathbf{1} = x_{\max}$$

2. Assume P is column-stochastic. Show using LP duality that $\exists y \in \mathbb{R}^n$ such that $Py = y$, $y \geq 0$ and $\mathbf{1}^T y = 1$.

Solution:

Consider the primal problem (in standard form):

$$\begin{aligned} \min_x \quad & -\mathbf{1}^T x \\ \text{s.t.} \quad & (P - I)x = 0 \\ & x \geq 0 \end{aligned}$$

and its dual

$$\begin{aligned} \max_y \quad & 0 \\ \text{s.t.} \quad & (P - I)^T y - \mathbf{1} \geq 0 \end{aligned}$$

Note that $x = 0$ is a feasible solution for the Primal problem.

Idea: show that the dual is infeasible, by using property from part 1.

Let $j = \arg \max y$ (i.e., j is the index of the largest element of the vector y). Then

$$((P - I)^T y)_j - 1 = (P^T y)_j - y_j - 1 \stackrel{\text{part 1}}{\leq} y_{\max} - y_j - 1 = -1 < 0$$

which violates the constraint in the dual problem.

This implies that the dual is infeasible and the primal is unbounded. Then

$$\exists x \text{ s.t. } x \geq 0, Px = x, \mathbf{1}x \geq 0$$

We then scale x to make $\mathbf{1}^T x = 1$, and conclude that

$$\exists x \text{ s.t. } x \geq 0, Px = x, \mathbf{1}^T x = 1.$$

Exercise 7

Consider the LP

$$\begin{array}{ll} \min_x & -2x_1 - 2x_2 \\ \text{s.t.} & x_1 - x_2 \leq 2 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{array}$$

1. Convert the problem into standard form, construct a BFS where $(x_1, x_2) = (0, 0)$

Solution:

We add slack variables (x_3, x_4) to convert the affine inequalities to equalities.

$$\begin{array}{ll} \min_x & -2x_1 - 2x_2 + 0x_3 + 0x_4 \\ \text{s.t.} & x_1 - x_2 + x_3 = 2 \\ & x_1 + x_2 + x_4 = 6 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{array}$$

The matrix A and the cost and rhs vectors of the LP in standard form are:

$$A = \begin{bmatrix} 1 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}, \quad c = \begin{pmatrix} -2 \\ -2 \\ 0 \\ 0 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$$

We note that the elements of the rhs vector b are all nonnegative. Then by setting $x_1, x_2 = 0$ we can determine the **initial** BFS for the LP in standard form:

$$x^{(0)} = \begin{pmatrix} x_1^{(0)} \\ x_2^{(0)} \\ x_3^{(0)} \\ x_4^{(0)} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \\ 6 \end{pmatrix}$$

We will use this BFS as our initial (or starting) point for the next part of this exercise.

2. Carry out 1 iteration of the Simplex method

Solution:

(a) We start by computing the basis and basic matrix A_B and cost vector c_B

$$B = \{3, 4\}, \quad A_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad c_B = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Now let's compute the reduced costs, dual variables

$$\bar{c} = c + A^T y \implies \bar{c} = \begin{pmatrix} -2 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \quad -A_B^T y = c_B \implies y = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Since there are negative reduced costs, the current BFS is not optimal. We choose the variables with the most negative reduced cost to enter the basis (this is x_1 , with reduced cost -2).

(b) Let's compute the feasible direction d , which consists of the non-basic vector d_N and the basic part d_B . Note that the direction d is very important because we will use it to move from the current BFS (which is a vertex) to a new BFS (which is a neighboring vertex) where the cost function decreases.

Since we have decided to have the non-basic variable x_1 to enter the , we will have $d_N^1 = 1$ and $d_N^j = 0, \forall j \neq 1$, that is

$$d_N = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

We can compute d_B by solving the system of linear equations

$$A_B d_B = -A_1 \implies \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} d_B = -\begin{pmatrix} 1 \\ 1 \end{pmatrix} \implies d_B = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

Hence the the feasible direction d is defined as

$$d = \begin{pmatrix} 1 \\ 0 \\ -1 \\ -1 \end{pmatrix}$$

(c) Now that we have computed the feasible direction, we need to compute the step-size. This is computed by the formula

$$\theta^* = \min\left\{\frac{x_i^{(0)}}{-d_i} \mid \text{for all } i \in B = \{3, 4\} \text{ and } d_i < 0\right\}$$

This gives us:

$$\theta^* = \min\left\{\frac{x_3^{(0)}}{-d_3}, \frac{x_4^{(0)}}{-d_4}\right\} = \min\left\{\frac{-2}{-1}, \frac{-6}{-1}\right\} = 2$$

Hence the next point is determined as follows:

$$x^{(1)} = x^{(0)} + \theta^* d = \begin{pmatrix} 0 \\ 0 \\ 2 \\ 6 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 0 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \\ 4 \end{pmatrix}$$

Note that the minimum ratio $-2/(-1)$ makes the first basic variable (x_3) to reach 0, as we increase the value of the non-basic variable x_1 (which enters the basis). Any value of $\theta^* > 2$ will make $x_3 < 0$ which will violate its nonnegativity constraint (we must always have $x_3 \geq 0$).

Note also that the minimum ratio identifies the variable that **exits** the basis. Hence the basic variable x_3 leaves the basis.

To summarize, we have x_1 entering the basis and x_3 exiting the basis. Hence the basis in the first iteration becomes

$$B = \{1, 4\}$$

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